S 2027

Sl. No. 2743648

SSLC EXAMINATION, MARCH - 2025

BIOLOGY

Time : 1½ Hours (English) Total Score : 40

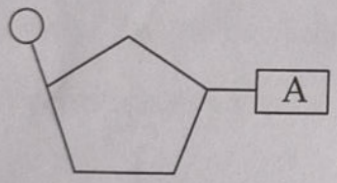
Instructions :

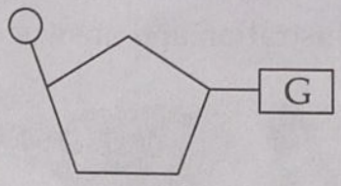
- The first **15** minutes is cool-off time.
- You may use the time to read the questions and plan your answers.
- Answer only on the basis of instructions and questions given.
- Consider score and time while answering.

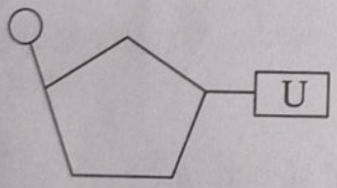
Score

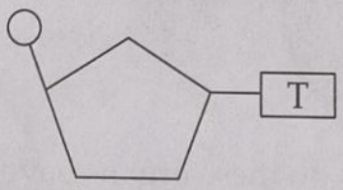
Answer any 5 questions from Q. No. 1 to 6. Each carries 1 score. **5x1=5**

1. Which is the main endocrine gland that controls metabolic activities ? **1**
   (a) Pancreas (b) Thyroid gland
   (c) Pineal gland (d) Adrenal gland

2. Which of the following nucleotides is found only in RNA ? **1**
   (a) 

   (b) 

   (c) 

   (d) 

3. "New species are formed by the inheritance of sudden changes that occur in genes". Which scientist proposed this explanation ? **1**
   (a) Charles Darwin (b) Lamarck
   (c) A.I. Oparin (d) Hugo de Vries

4. Which of the following activity is **correct** while viewing nearby objects ? **1**
   (i) The ciliary muscles contract
   (ii) The curvature of the lens decreases.
   (iii) The ligaments stretch.
   (iv) The focal length decreases.
   (a) (i) and (ii) are correct (b) (i), (ii) and (iii) are correct
   (c) (i) and (iv) are correct (d) (ii), (iii) and (iv) are correct

Score

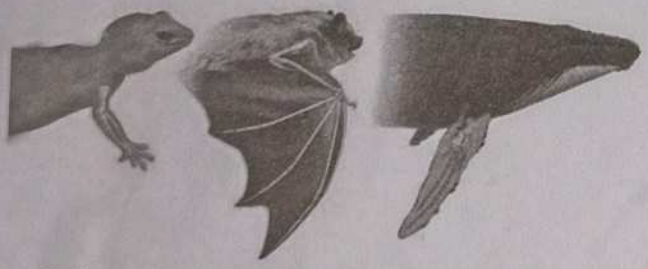
5. Correct the mistakes, if any in the underlined part of the given statements. 1
   (a) BCG vaccine is used to prevent Tuberculosis.
   (b) AB positive is the blood group that has the D antigen in addition to A and B antigens.
   (c) Fibrin is the enzyme that converts prothrombin to thrombin in the process of blood clotting.

6. Some diseases and proteins produced through Genetic engineering for their treatment are given. Choose the correct pair. 1
   (a) Insulin - Growth disorders
   (b) Interferons - Viral diseases
   (c) Endorphins - Diabetes
   (d) Somatotropin - Pain

Answer any 6 questions from Q. No. 7 to 13. Each carries 2 scores. 6x2=12

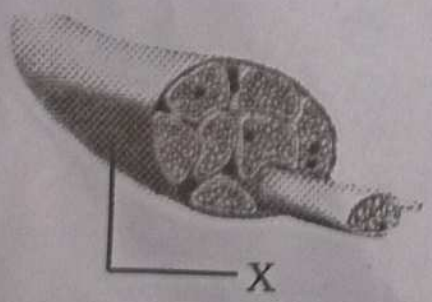
7. Analyse the given statements **'A'** and **'B'** and answer the following questions.
   A : The defects in genes may cause deformities in the sequencing of amino acids which are the building blocks of haemoglobin.
   B : The genes that control the synthesis of proteins for blood clotting become defective.
   (a) Identify the diseases indicated by **'A'** and **'B'** ? 1
   (b) How is a temporary relief brought about for the disease indicated by **'B'** ? 1

8. Observe the given illustration and answer the questions.



   (a) What similarity do these organs have in their internal structure ? 1
   (b) How does the comparative study of these organs provide evidence for evolution ? 1

9. Observe the cross-section of a nerve and write answers to the questions.



   (a) What is the function of the part indicated by 'X' ? 1
   (b) How do sensory nerves and motor nerves differ in function ? 1

Score

10. Analyse the newspaper report given below and answer the questions.

> The spread of the HIV increases through drug injection.
> Most of the infected people are adolescents...!

(a) How does drug injection lead to the spread of the HIV ? 1

(b) How does the condition of AIDS become fatal ? 1

11. Explain the role of each of the following activities in defense mechanism.

(a) Lysosomes combine with membrane sac during the process of phagocytosis. 1

(b) Chemicals produced by white blood cells increase body temperature. 1

12. Analyse each of the given situations and explain how variations are formed in the offspring.

(a) During the initial phase of meiosis, chromosomes pair. 1

(b) Parents' chromosomes reach their offspring through gametes. 1

13. The main parts and functions of brain are given in the box. Arrange them in the following table suitably. 2

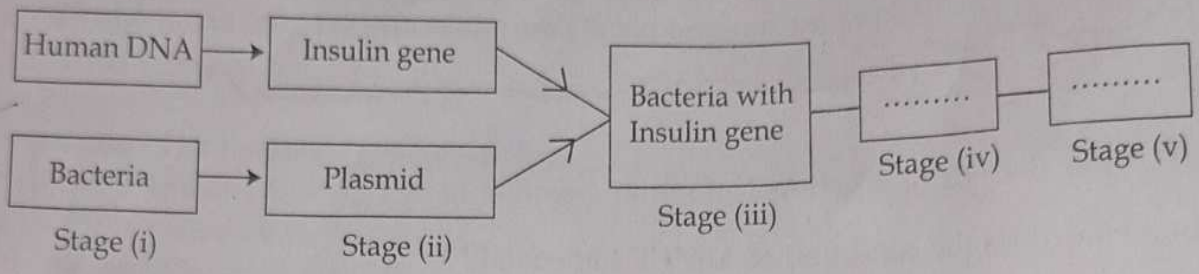
> Thalamus, Controls voluntary movements, Cerebellum, Acts as relay station of impulses, Hypothalamus, Maintains equilibrium of the body, Cerebrum, Maintains homeostasis

Part	Function
•	•
•	•
•	•
•	•

Score
5x3=15

Answer any 5 questions from Q. No. 14 to 20. Each carries 3 scores.

14. Analyse the illustration indicating the stages of Insulin production through Genetic Engineering and answer the following questions.



(a) Name the enzymes used in this process. 1

(b) What is the importance of plasmid in this process ? 1

(c) Complete the stages indicated by (iv) and (v). 1

15. The normal level of blood glucose is 70-110 mg/100 ml.

(a) What are the two reasons that cause glucose to rise above this level and be excreted in urine ? 2

(b) Mention one function of insulin and one function of glucagon in regulating blood glucose levels. 1

16. Answer the following questions related to an experiment in which a tall pea plant (TT) is crossed with a dwarf pea plant (tt).

(a) Write the genetic structure and character of the first generation. 1

(b) Illustrate the experiment using symbols in which the second generation is produced. 2

17. Analyse the table showing defects or diseases and their causes, then answer the following questions.

Defect/Disease	Cause
Xerophthalmia	..... (i) .....
..... (ii)	Defects in cone cells
..... (iii)	Reabsorption of aqueous humor does not occur
Night blindness	..... (iv) .....

(a) Identify and write (i), (ii), (iii) and (iv). 2

(b) What is the treatment for the disease indicated by (iii) ? 1

Score

18. Observe the illustration regarding different types of synapses and answer the questions.

A : Neuron — Neuron

B : Neuron — (i)

C : Neuron — (ii)

(a) Identify and write the parts that are indicated by (i) and (ii). 1

(b) What is the role of neurotransmitters in the transmission of impulses across the synapse ? 1

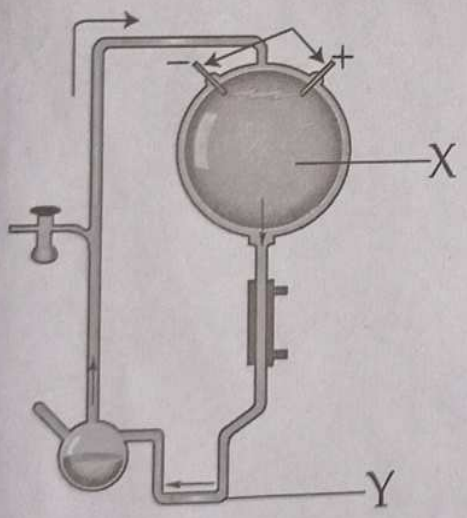
(c) What are the two main functions of synapses ? 1

19. The questions raised in an open discussion organized by the Science Club regarding lifestyle diseases are given below. Write your explanation for the questions.

(a) What is meant by lifestyle diseases ? 1

(b) What are the healthy habits that can be adopted to avoid such diseases ? 2

20. Observe the experimental setup given below and answer the questions.



(a) What are the chemicals used in the part labelled 'X' in the experiment ? 1

(b) What substance was obtained by examining the precipitates in the part labelled 'Y' ? 1

(c) What is the evolutionary significance of the synthesis of more such compounds in similar experiments ? 1

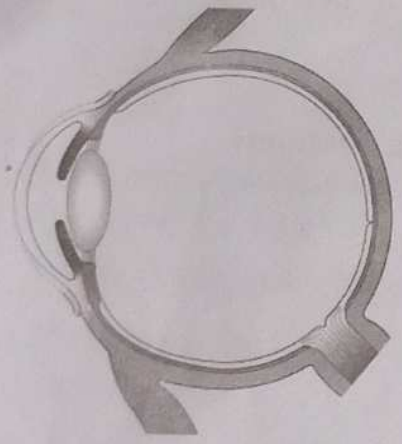
Score

Answer any 2 questions from Q. No. 21 to 23. Each carries 4 scores. 2x4=8

21. Arrange columns **B** and **C** according to column **A**. 4

A - Gland	B - Hormone	C - Function
Pituitary	Progesterone	Slows down the action of defense cells
Adrenal gland	Releasing hormone	Production of milk
Ovary	Prolactin	Controls secondary sexual characters
Hypothalamus	Cortisol	Implantation of embryo in the uterus.
	Calcitonin	Controls the production of tropic hormones

22. Redraw the diagram, identify and label the parts with their names.



To redraw the diagram 1

(a) The part that allows light rays to enter the eye. 1

(b) The muscles that adjust the curvature of the lens. 1

(c) The part that adjusts its size according to the intensity of light. 1

23. Answer the questions given below regarding diagnosis and treatment in Modern medicine.

(a) What are the uses of EEG and ECG ? 2

(b) What are the uses of antibiotics ? What are the side effects of their regular use ? 2

- o O o -